

Advanced Bayesian Learning

Mixture models and Bayesian nonparametrics

Lecture 3

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Topic overview

- Reminder: Multinomial data - Dirichlet prior
- Bayesian histograms
- The Dirichlet process
- Beyond DP: Pitman-Yor and Probit stick-breaking
- Finite mixture models
- Dirichlet process mixtures
- MCMC for Dirichlet process mixtures

The Dirichlet distribution

- $\theta \sim \text{Dirichlet}(\alpha)$ with density ($\alpha = (\alpha_1, \dots, \alpha_K)$)

$$p(\theta_1, \theta_2, \dots, \theta_K) \propto \prod_{j=1}^K \theta_j^{\alpha_j-1}.$$

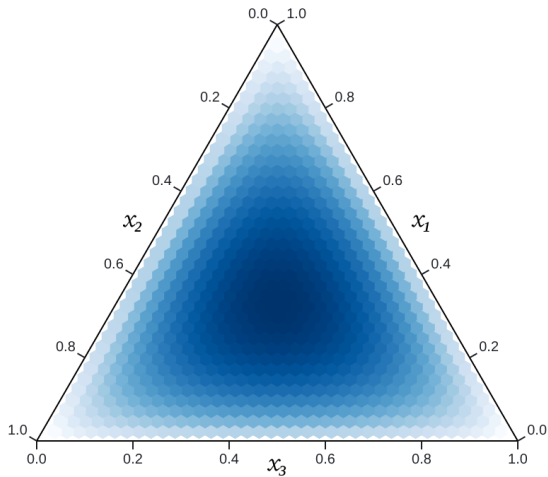
- Define $\alpha_0 = \sum_{j=1}^K \alpha_j$ and $\bar{\alpha} = \alpha / \alpha_0$.

- **Expected value** and **variance** of $\text{Dirichlet}(\alpha_1, \dots, \alpha_K)$

$$\mathbb{E}(\theta_j) = \frac{\alpha_j}{\alpha_0} = \bar{\alpha}_j \qquad \mathbb{V}(\theta_j) = \frac{\mathbb{E}(\theta_j) [1 - \mathbb{E}(\theta_j)]}{1 + \alpha_0}$$

- Note that α_0 is a **precision** parameter (large α_0 , low variance).
- We can also write $\theta \sim \text{Dirichlet}(\alpha_0 \bar{\alpha})$.

Interactive - Dirichlet distribution



Conjugate analysis for multinomial data

■ **Data:** $\mathbf{n} = (n_1, \dots, n_K)$, where $n_k = \#$ items in category k .

■ **Prior**

$$\boldsymbol{\theta} \sim \text{Dirichlet}(\boldsymbol{\alpha})$$

■ **Likelihood**

$$p(n_1, n_2, \dots, n_K | \theta_1, \theta_2, \dots, \theta_K) \propto \prod_{k=1}^K \theta_k^{n_k}$$

■ **Posterior**

$$\boldsymbol{\theta} | \mathbf{n} \sim \text{Dirichlet}(n_1 + \alpha_1, \dots, n_K + \alpha_K) = \text{Dirichlet}(\mathbf{n} + \boldsymbol{\alpha})$$

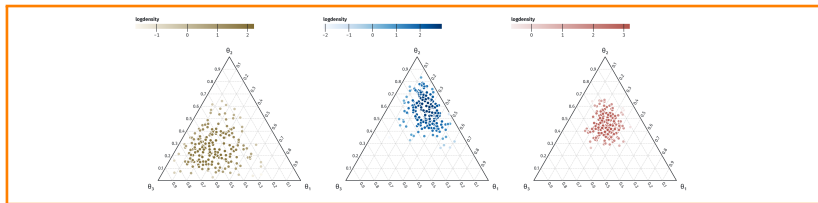
■ **Posterior mean**

$$\mathbb{E}(\boldsymbol{\theta} | \mathbf{n}) = \frac{\mathbf{n} + \boldsymbol{\alpha}}{n + \alpha_0}$$

■ **Alternatively**

$$\boldsymbol{\theta} | \mathbf{n} \sim \text{Dirichlet}\left((n + \alpha_0) \frac{\mathbf{n} + \boldsymbol{\alpha}}{n + \alpha_0}\right)$$

Interactive - Multinomial model



Bayesian histograms

- **Partition** the data space into disjoint intervals (**bins**): $B_1, B_2, \dots, B_K$. Width of B_k is Δ_k .
- The usual histogram estimate

$$\hat{p}(x) = \frac{n_k/n}{\Delta_k}$$

- **Probability model** for **histograms**

$$p(x) = \sum_{k=1}^K \frac{\omega_k}{\Delta_k} 1(x \in B_k)$$

- n_k = number of obs in B_k . Multinomial data.
- **Prior** on $\omega = (\omega_1, \dots, \omega_K)$

$$\omega \sim \text{Dirichlet}(\alpha) = \text{Dirichlet}(\alpha_0 \bar{\alpha})$$

- **Posterior**

$$\omega | \mathbf{n} \sim \text{Dirichlet}\left((n + \alpha_0) \frac{\mathbf{n} + \boldsymbol{\alpha}}{n + \alpha_0}\right)$$

- total counts = data counts + prior counts.

Bayesian histograms

■ Prior

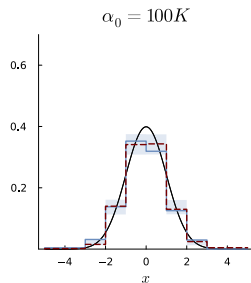
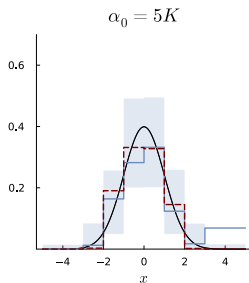
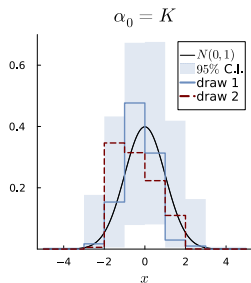
$$\text{Dirichlet}(\alpha_0 \bar{\alpha})$$

- Specify prior mean $\bar{\alpha}$ from a **base distribution** P_0 :

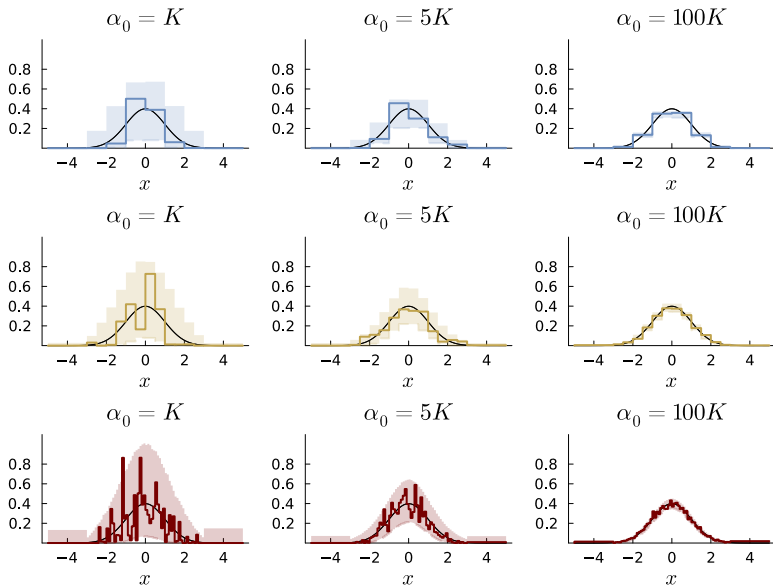
$$\bar{\alpha}_k = P_0(B_k) = \Pr(x \in B_k).$$

- Specify the strength of prior beliefs by α_0 .
- Properties of Dirichlet prior:
 - ▶ **easy computations**
 - ▶ **easy to specify hyperparameter** $\bar{\alpha}$ and α_0 .
 - ▶ **no smoothness**: adjacent bin are negatively correlated.
 - ▶ **sensitive** to the choice of **bins**.

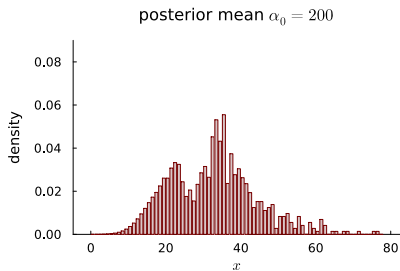
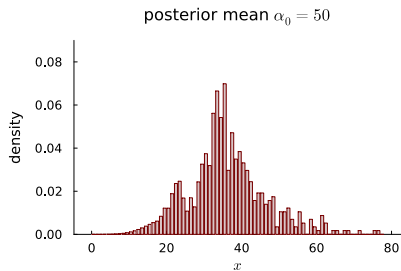
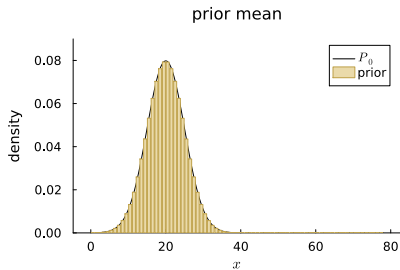
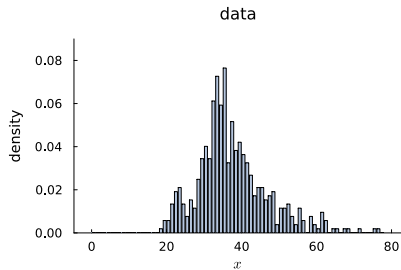
Simulation from histogram prior



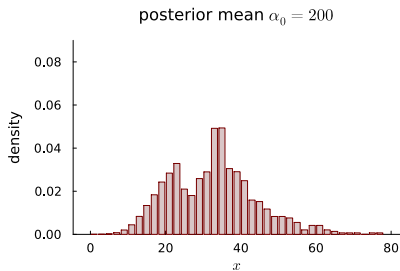
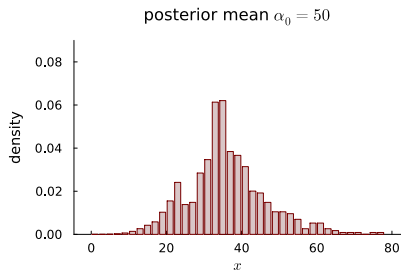
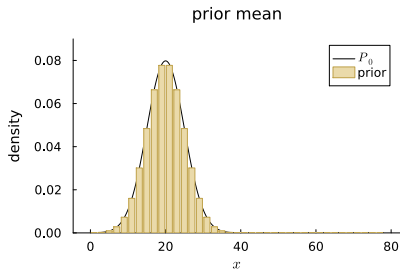
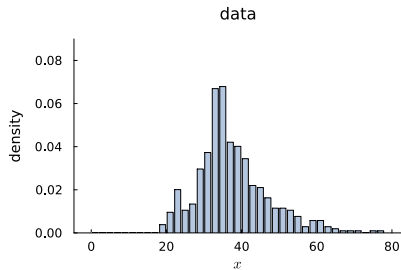
Simulation from histogram prior - bin size



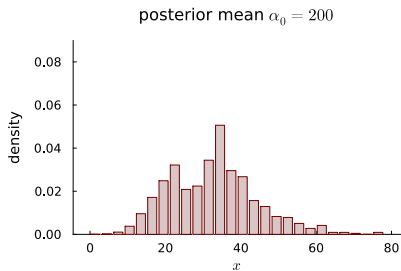
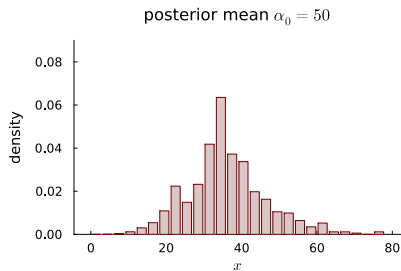
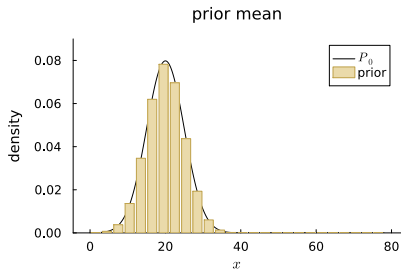
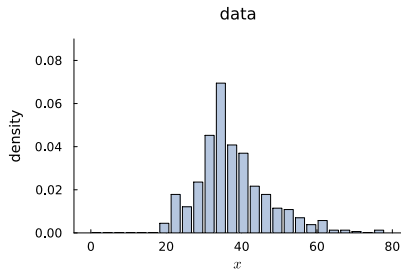
Pike length data - Bayesian histogram with $\Delta = 1$



Pike length data - Bayesian histogram with $\Delta = 2$



Pike length data - Bayesian histogram with $\Delta = 3$



The Dirichlet process

- Let $B_1, B_2, \dots, B_k$ be a partition of the outcome space Ω .
- $P(B_1), \dots, P(B_k)$ denotes the distribution over the partition.
- Dirichlet distribution is a **distribution over distributions**:

$$(P(B_1), \dots, P(B_k)) \sim \text{Dirichlet}(\alpha_0 P_0(B_1), \dots, \alpha_0 P_0(B_k))$$

where P_0 is a fixed probability measure (e.g. $N(0, 1)$).

- Dirichlet is **closed under summation or splitting** of bins.
 $\Rightarrow$ consistent definition of a **stochastic process**. c.f. GPs.
- A random probability measure P follows a **Dirichlet process** $P \sim \text{DP}(\alpha_0 \cdot P_0)$ with **base measure** P_0 iff

$$(P(B_1), \dots, P(B_k)) \sim \text{Dirichlet}(\alpha_0 P_0(B_1), \dots, \alpha_0 P_0(B_k))$$

for any finite measurable partition $B_1, \dots, B_k$.

The Dirichlet process - properties

- If $P \sim \text{DP}(\alpha_0 P_0)$ then

$$P(B) \sim \text{Beta}[\alpha_0 P_0(B), \alpha_0 (1 - P_0(B))], \text{ for any } B \in \mathcal{B}$$

$$\mathbb{E}[P(B)] = P_0(B)$$

$$\mathbb{V}[P(B)] = P_0(B) [1 - P_0(B)] / (1 + \alpha_0)$$

- **Model**

$$y_i | P \stackrel{\text{iid}}{\sim} P, \text{ for } i = 1, \dots, n$$

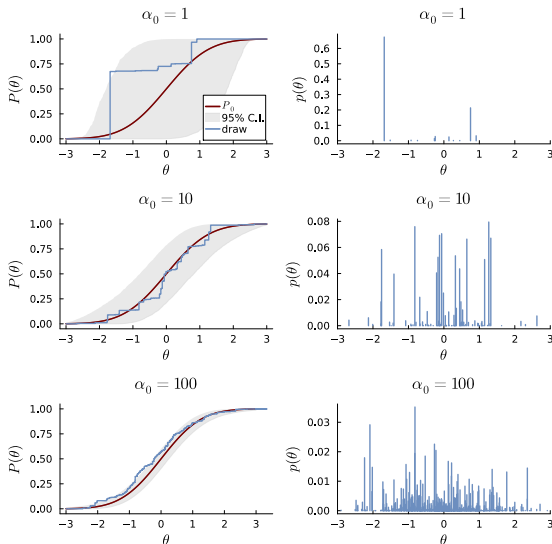
- **Prior**

$$P \sim \text{DP}(\alpha_0 P_0)$$

- **Posterior** for a **finite partition**, $P(B_1), \dots, P(B_k) | y$ is

$$\text{Dirichlet} \left(\alpha_0 P_0(B_1) + \sum_{i=1}^n 1_{y_i \in B_1}, \dots, \alpha_0 P_0(B_k) + \sum_{i=1}^n 1_{y_i \in B_k} \right)$$

Dirichlet process realizations



The Dirichlet process - properties

- **Posterior** for the unknown probability distribution P

$$P|y_1, \dots, y_n \sim \text{DP} \left(\alpha_0 P_0 + \sum_{i=1}^n \delta_{y_i} \right)$$

- Since

$$P(B) \sim \text{Beta} \left(\alpha_0 P_0(B) + \sum_{i=1}^n 1_{y_i \in B}, \alpha_0 (1 - P_0(B)) + \sum_{i=1}^n 1_{y_i \in B^c} \right)$$

so

$$\mathbb{E}(P(B)|y_1, \dots, y_n) = \left(\frac{\alpha_0}{\alpha_0 + n} \right) P_0(B) + \left(\frac{n}{\alpha_0 + n} \right) \sum_{i=1}^n \frac{1}{n} \delta_{y_i}(B)$$

Estimating a distribution function with a DP prior

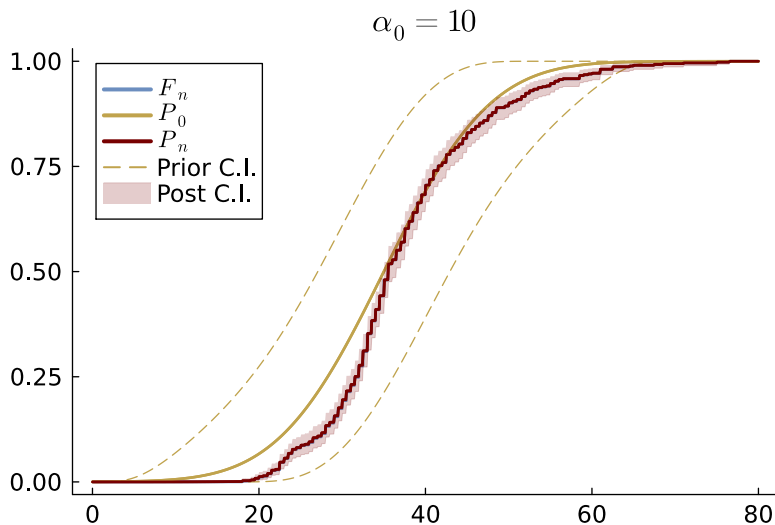
- If $B = (-\infty, y]$ then

$$E(F(y)|y_1, \dots, y_n) = \left(\frac{\alpha_0}{\alpha_0 + n}\right) F_0(y) + \left(\frac{n}{\alpha_0 + n}\right) F_n(y)$$

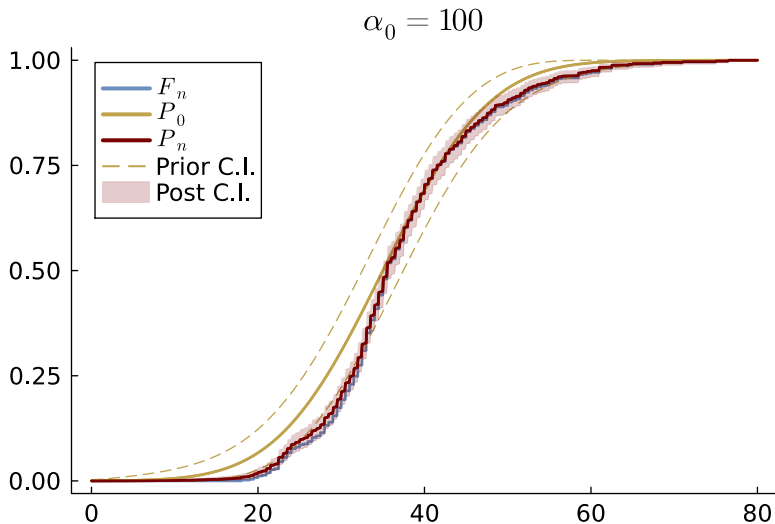
- ▶ $F(y)$ is the unknown cdf
- ▶ $F_0(y)$ is the cdf from P_0
- ▶ $F_n(y) = \frac{1}{n} \sum 1_{y_i \leq y}$ is the empirical cdf

- $F(\cdot)$ is **discrete with probability one** in the DP posterior.
- **Realisations from a DP are discrete with probability one.**
 - ▶ Clearly a bad property for continuous data ...
 - ▶ But very useful for clustering (mixture models).

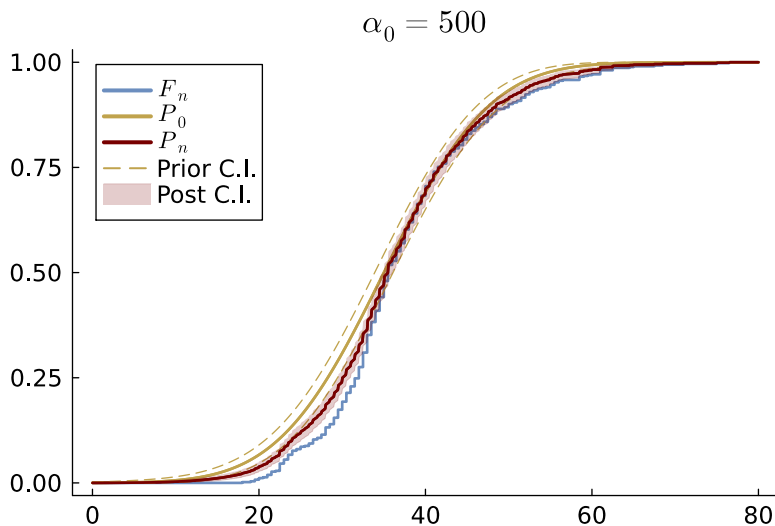
Fish length data $\alpha_0 = 10$



Fish length data $\alpha_0 = 100$



Fish length data $\alpha_0 = 500$



Stick-breaking characterization of the DP

- $P \sim DP(\alpha_0 P_0) \equiv$ infinite mixture of point masses

$$P(\cdot) = \sum_{h=1}^{\infty} \pi_h \delta_{\theta_h}$$

$$\pi_h = V_h \prod_{\ell < h} (1 - V_\ell)$$

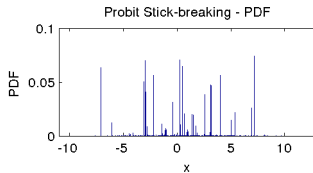
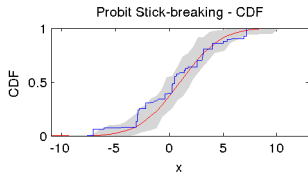
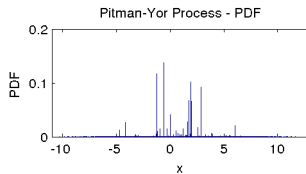
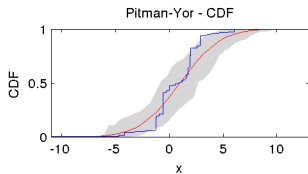
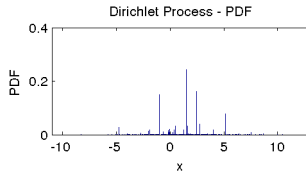
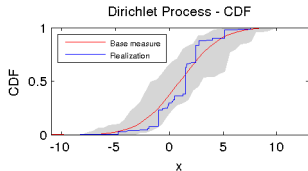
$$V_h \stackrel{\text{iid}}{\sim} \text{Beta}(1, \alpha_0)$$

$$\theta_h \stackrel{\text{iid}}{\sim} P_0$$

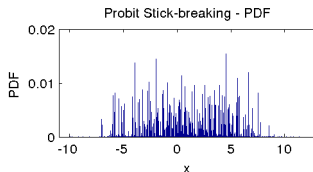
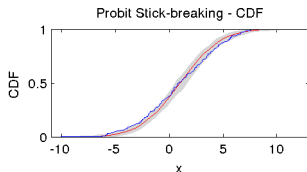
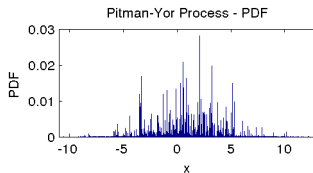
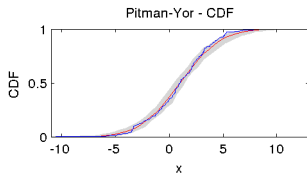
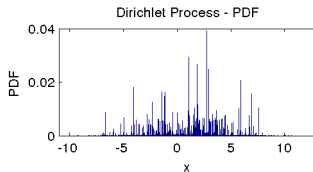
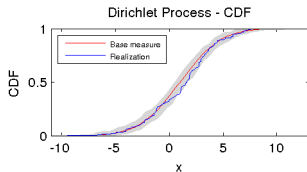
- Alternative notation for $P \sim DP(\alpha_0 P_0)$:

$$\pi = (\pi_1, \pi_2, \dots) \sim \text{Stick}(\alpha_0) \text{ and } \theta_h \sim P_0$$

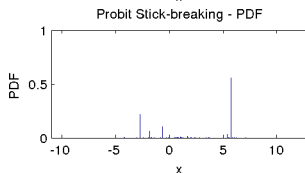
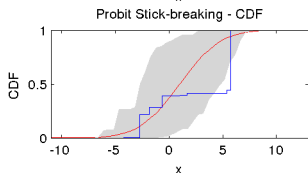
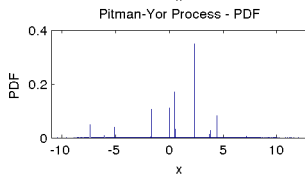
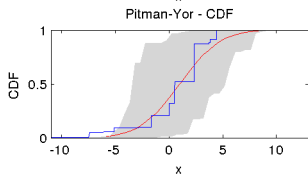
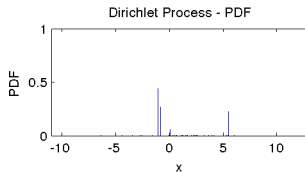
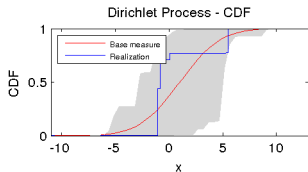
Simulating stick-breaking $\alpha_0 = 10, P_0 = N(1, 3^2)$



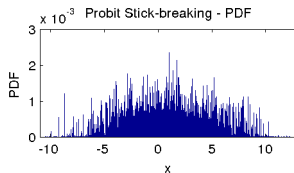
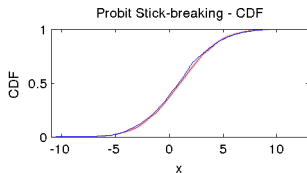
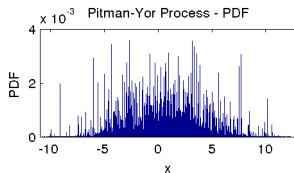
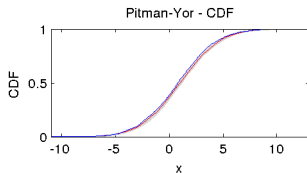
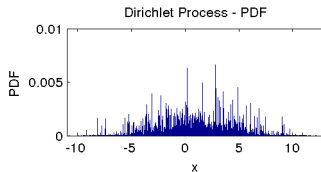
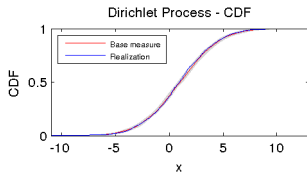
Simulating stick-breaking $\alpha_0 = 100$, $P_0 = N(1, 3^2)$



Simulating stick-breaking $\alpha_0 = 1, P_0 = N(1, 3^2)$



Simulating stick-breaking $\alpha_0 = 1000$, $P_0 = N(1, 3^2)$



Beyond DP - Pitman-Yor and Probit sticks

- **Pitman-Yor process** with parameters P_0 , $0 \leq a < 1$ and $b > -a$:

$$P(\cdot) = \sum_{h=1}^{\infty} \pi_h \delta_{\theta_h} \quad \theta_h \stackrel{iid}{\sim} P_0$$

$$\pi_h = V_h \prod_{\ell < h} (1 - V_\ell)$$

$$V_h \stackrel{iid}{\sim} \text{Beta}(1 - a, b + ha)$$

- **Probit stick-breaking** with parameters μ and σ :

$$P(\cdot) = \sum_{h=1}^{\infty} \pi_h \delta_{\theta_h} \quad \theta_h \stackrel{iid}{\sim} P_0$$

$$\pi_h = V_h \prod_{\ell < h} (1 - V_\ell)$$

$$V_h = \Phi(x_h), \quad \text{where } x_h \stackrel{iid}{\sim} N(\mu, \sigma^2)$$